

LESSON 7-6 • HOMEWORK PACKET • SHEET 1 OF 2

Distance Between Points on a Grid

Maryland Standard 6.NS.C.8 • Learning target – I can find the distance between two points that share a coordinate.

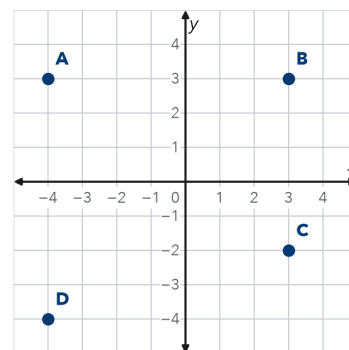
REMEMBER

- **Same x-coordinate:** the points are on a vertical line. Use the y-coordinates.
- **Same y-coordinate:** the points are on a horizontal line. Use the x-coordinates.
- Opposite signs: **add** the absolute values. Same signs: **subtract** them.

WORKED EXAMPLEFind the distance from $(3, -4)$ to $(3, 5)$.**Step 1** Same x, so use the y-coordinates -4 and 5 .**Step 2** Opposite signs, so add: $4 + 5 = 9$.**Answer 9 units****PRACTICE***show your work • label units*

1. Use the grid. Find each distance.

- a. A to B _____ units
b. B to C _____ units
c. A to D _____ units



2. Find the distance between each pair of points.

- a. $(5, 6)$ and $(5, -2)$ _____
b. $(-7, 1)$ and $(-2, 1)$ _____
c. $(-3, -4)$ and $(6, -4)$ _____
d. $(0, -8)$ and $(0, 3)$ _____
e. $(-9, -2)$ and $(-9, -7)$ _____
f. $(2.5, 3)$ and $(-4, 3)$ _____
g. $(-1, -6)$ and $(-1, 4)$ _____
h. $(-8, 5)$ and $(-3.5, 5)$ _____

3. **City Blocks.** Kim walks straight from $(-6, -3)$ to $(-6, 5)$. Each unit is 1 block. How far does she walk?

Answer _____ blocks

4. **Missing Point.** Point P is at $(2, -3)$. Point Q is 6 units to the left of P. Name point Q.

Answer _____

5. **Find the Error.** Eli says the distance from $(1, -6)$ to $(1, -2)$ is 8 because $6 + 2 = 8$. Explain his mistake and find the correct distance.

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME With someone at home, count the blocks between two places on one street.

LESSON 7-6 • HOMEWORK PACKET • SHEET 2 OF 2**Comparing Distances on the Plane**

Maryland Standard 6.NS.C.8 • Learning target – I can decide whether to add or subtract to find a distance, then compare two distances.

REMEMBER

- Find **each** distance first. Then compare.
- Different signs: **add**. Same sign: **subtract**.
- A distance is never negative.

WORKED EXAMPLE

Which is longer: $(-2, 4)$ to $(5, 4)$, or $(1, -3)$ to $(1, 5)$?

Step 1 First: same y , use x : $2 + 5 = 7$ units.

Step 2 Second: same x , use y : $3 + 5 = 8$ units. $8 > 7$.

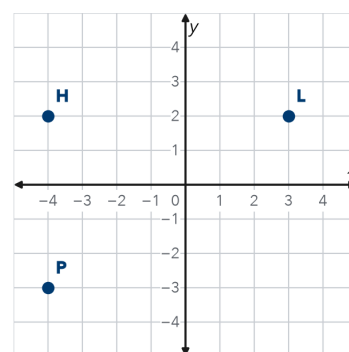
Answer the second one

PRACTICE

show your work • label units

- 1. Neighborhood.** Each unit is 1 block. Which is closer to Home: the Library or the Park?

- a. H to L _____ blocks
- b. H to P _____ blocks
- c. Which is closer? By how many blocks? _____



- 2.** Find each distance.

- a. $(-8, 1)$ to $(-3, 1)$ _____
- b. $(4, -6)$ to $(4, 7)$ _____
- c. $(-6, -7)$ to $(-6, -1)$ _____
- d. $(-2.5, 3)$ to $(4.5, 3)$ _____

- 3. Runners.** Ana runs from $(-5, -1)$ to $(6, -1)$. Jay runs from $(2, 4)$ to $(2, -6)$. Who runs farther, and by how many units?

Answer _____

- 4. Reasoning.** For $(-3, 7)$ to $(-3, -2)$, do you add or subtract? Explain, then find the distance.

TEST PRACTICE

choose the best answer

- 5.** Segment JK runs from $(-4, 2)$ to $(6, 2)$. Segment MN runs from $(1, -3)$ to $(1, 5)$. Which is longer?
- A** JK, by 2 units **B** MN, by 2 units **C** neither **D** JK, by 8 units
- 6.** Point A is at $(-2, -5)$. Point B is at $(-2, 3)$. What is the distance from A to B?
- A** 2 units **B** 3 units **C** 8 units **D** -8 units

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME With someone at home, pick two places on one street. Which is closer to your home? By how many blocks?